VOL. I · NO. 8 MONDAYS & THURSDAYS

THE LAUNCH REPORT

Rocket launches, reentries and the business of spaceflight, worldwide

THURSDAY, NOVEMBER 5, 2026 ALL TIMES EASTERN STANDARD TIME (EST)

THE WEEK IN LAUNCH

Electron's 99th and a Spacesail cluster open November

Two orbital launches flew worldwide since Monday's issue, both successful. Rocket Lab's Electron lofted a Japanese radar satellite on its 99th flight Monday morning from Mahia, and China's Kinetica 2 put 18 Spacesail communications satellites into polar orbit Monday evening from Jiuquan. That brings 2026 to 255 orbital launch attempts, 248 of them successful, according to GCAT plus Monday's flights from Launch Library 2.

USSF-432 has slipped again, to Friday night in the FAA plan, and Starlink 15-21 is off the near-term board. NASA's Neil Gehrels Swift Observatory reentered Wednesday over Indonesia after nearly 22 years, and Canadian and NASA astronauts were due outside the International Space Station this morning to install the final roll-out solar array.

I · RECENT LAUNCHES & RETURNS

Electron hits 99; Kinetica 2 flies Spacesail's polar group

Rocket Lab's 18th orbital launch of the year and China's second Kinetica 2 both reached orbit.

Electron: "The Sea God Surveys"

"We've officially launched Electron 99 times," Rocket Lab posted after the flight. "It took us seven years to reach 50. Now, just two years later, we're closing in on 100." Liftoff was at 11:41 a.m. EST Monday (16:41 UTC) from Launch Complex 1A on New Zealand's Mahia Peninsula, the opening of a two-hour window (NASASpaceflight; Spaceflight Now). The rocket put QPS-SAR-20, also called WADATSUMI-II, into low Earth orbit for Japan's Institute for Q-shu Pioneers of Space on the company's 10th dedicated Electron mission and first attempt of the day (Rocket Lab).

"Mission success! Repeat customer @QPS_Inc has made contact with its latest QPS-SAR satellite following their tenth ride on Electron," the company confirmed later Monday. It was Electron's 17th flight of 2026 and Rocket Lab's 18th orbital launch of the year per GCAT.

Kinetica 2: SpaceSail Polar Group #17

CAS Space's Kinetica 2 lifted off at about 7:15 p.m. EST Monday (00:15 UTC Tuesday) from Launch Area 140 at Jiuquan and reached orbit with 18 low-Earth-orbit communications satellites for Shanghai Spacesail Technologies' G60 constellation, according to Launch Library 2. The satellites carry Ku-, Q- and V-band payloads; Spacesail plans 1,296 satellites by 2027. It was the second flight of Kinetica 2 after its March debut (NASASpaceflight). No other orbital launches took place between Monday's pair and Thursday morning (Launch Library 2).

Three Vandenberg Falcons and Vulcan Leo 1 fill mid-November

Every orbital launch with a date between now and November 30. Times are EST with Coordinated Universal Time (UTC) below; NET means no earlier than.

DATE / TIME	VEHICLE	PROVIDER	MISSION / PAYLOAD	SITE	ORBIT	STATUS	SOURCES
Thu Nov 5, 11:00 a.m. 16:00 UTC	Long March 2C	CASC	Payload not announced	Launch Complex 3, Xichang, China	—	**TENTATIVE**	Launch Library 2
Fri Nov 6, 6:35 p.m. 23:35 UTC	Ceres-2	Galactic Energy	Payload not announced; return to flight after January failure	Launch Area 95A, Jiuquan, China	—	**TENTATIVE**	Launch Library 2
Fri Nov 6, 7:46 p.m. 00:46 UTC Nov 7	Falcon 9	SpaceX	USSF-432, a classified U.S. Space Force mission (FAA: "R-4")	SLC-4E, Vandenberg, Calif.	LEO	**CONFIRMED**	FAA; Launch Library 2
Sun Nov 8, 3:25 p.m. 20:25 UTC	Long March 12	CASC	Nine SatNet low-Earth-orbit communications satellites (group not confirmed)	Commercial LC-2, Wenchang, China	LEO	**TENTATIVE**	Launch Library 2
Mon Nov 9, 10:03 p.m. 03:03 UTC Nov 10	Falcon 9	SpaceX	USSF-627, a classified U.S. Space Force mission (FAA: "R-5")	SLC-4E, Vandenberg, Calif.	LEO	**CONFIRMED**	FAA; Launch Library 2
Tue Nov 10, 1:00 a.m. 06:00 UTC	Mir (name provisional)	Agency for Defense Development, South Korea	Demonstration flight, first full orbital launch of the military small-satellite rocket	Offshore launch platform	LEO	**TENTATIVE**	Launch Library 2
Wed Nov 11, 6:40 a.m. 11:40 UTC	Long March 2D	CASC	Payload not announced	Launch Area 94, Jiuquan, China	—	**TENTATIVE**	Launch Library 2
Thu Nov 12, 11:23 p.m. 04:23 UTC Nov 13	Falcon 9	SpaceX	USSF-317, a classified U.S. Space Force mission	SLC-4E, Vandenberg, Calif.	LEO	**CONFIRMED**	FAA; Launch Library 2
Fri Nov 13, 12:54 p.m. 17:54 UTC	Vulcan VC6L	United Launch Alliance	Vulcan Leo 1 (LV-01): 40 Amazon Leo broadband satellites	SLC-41, Cape Canaveral, Fla.	LEO	**CONFIRMED**	ULA; Launch Library 2
NET Fri Nov 13	Electron	Rocket Lab	"The Mountain God Rises": iQPS radar imaging satellite	Launch Complex 1, Mahia, New Zealand	LEO	**PRELIMINARY**	Launch Library 2
Fri Nov 20, 6:00 a.m. 11:00 UTC	Long March 8A	CASC	Nine SatNet low-Earth-orbit communications satellites (group not confirmed)	Commercial LC-1, Wenchang, China	LEO	**TENTATIVE**	Launch Library 2
NET Sun Nov 22	Ariane 64	Arianespace	VA271: 36 Amazon Leo broadband satellites, 10th Ariane 6	ELA-4, Kourou, French Guiana	LEO	**PRELIMINARY**	Launch Library 2
Tue Nov 24, 12:03 a.m. 05:03 UTC	Soyuz-2.1b	Roscosmos	Progress MS-36 cargo ship to the International Space Station	Site 31/6, Baikonur, Kazakhstan	LEO	**TENTATIVE**	Launch Library 2
Tue Nov 24, 2:00 a.m. 07:00 UTC	Long March 12	CASC	Nine SatNet low-Earth-orbit communications satellites (group not confirmed)	Commercial LC-2, Wenchang, China	LEO	**TENTATIVE**	Launch Library 2
NET Tue Nov 24	Long March 2F	CASC	Shenzhou 24 crew to the Tiangong station	Launch Area 91, Jiuquan, China	LEO	**PRELIMINARY**	Launch Library 2
Thu Nov 26, 3:00 p.m. 20:00 UTC	Spectrum	Isar Aerospace	Third test flight, carrying several satellites including R-Space's AT-Astra	Andøya Spaceport, Norway	SSO	**TENTATIVE**	Launch Library 2
Sun Nov 29, 10:49 a.m. 15:49 UTC	Falcon 9	SpaceX	Northrop Grumman's Cygnus NG-25 cargo ship to the space station	SLC-40, Cape Canaveral, Fla.	LEO	**CONFIRMED**	Launch Library 2

CONFIRMED date and time announced by the launch provider, its customer or the FAA. **TENTATIVE** specific date in published schedules or airspace and maritime notices, not yet confirmed by the operator. **PRELIMINARY** target range, or no-earlier-than date or month only.

Abbreviations: SLC, Space Launch Complex; SLC-4E, Space Launch Complex 4 East; ELA-4, Ariane Launch Area 4; LEO, low Earth orbit; SSO, Sun-synchronous orbit; CASC, China Aerospace Science and Technology Corp.; iQPS, Institute for Q-shu Pioneers of Space, a Japanese radar imaging company; USSF, U.S. Space Force.

Date changes

USSF-432. Tuesday at 8:28 p.m.; now Friday at 7:46 p.m., the FAA primary window for "R-4" (FAA; Launch Library 2). No reason given.

Starlink 15-21. Thursday at 9:21 p.m.; now NET November 30 with no specific time, and absent from Thursday's FAA plan (Launch Library 2; FAA). No reason given.

Long March 12. Thursday at 4:15 p.m.; now Sunday at 3:25 p.m. (Launch Library 2). No reason given.

Mir demonstration. Wednesday, November 4; now Tuesday, November

10, at 1:00 a.m. (Launch Library 2). No reason given.

USSF-627. Listed Monday as "R-5" without a mission name; now identified as USSF-627 at the same Monday 10:03 p.m. window (Launch Library 2; FAA).

Vulcan Leo 1. Friday, November 13, date only; now Friday at 12:54 p.m. (Launch Library 2). ULA still lists the day "pending range approval" (ULA).

Electron "The Mountain God Rises." NET November 14; now NET November 13 (Launch Library 2). No reason given.

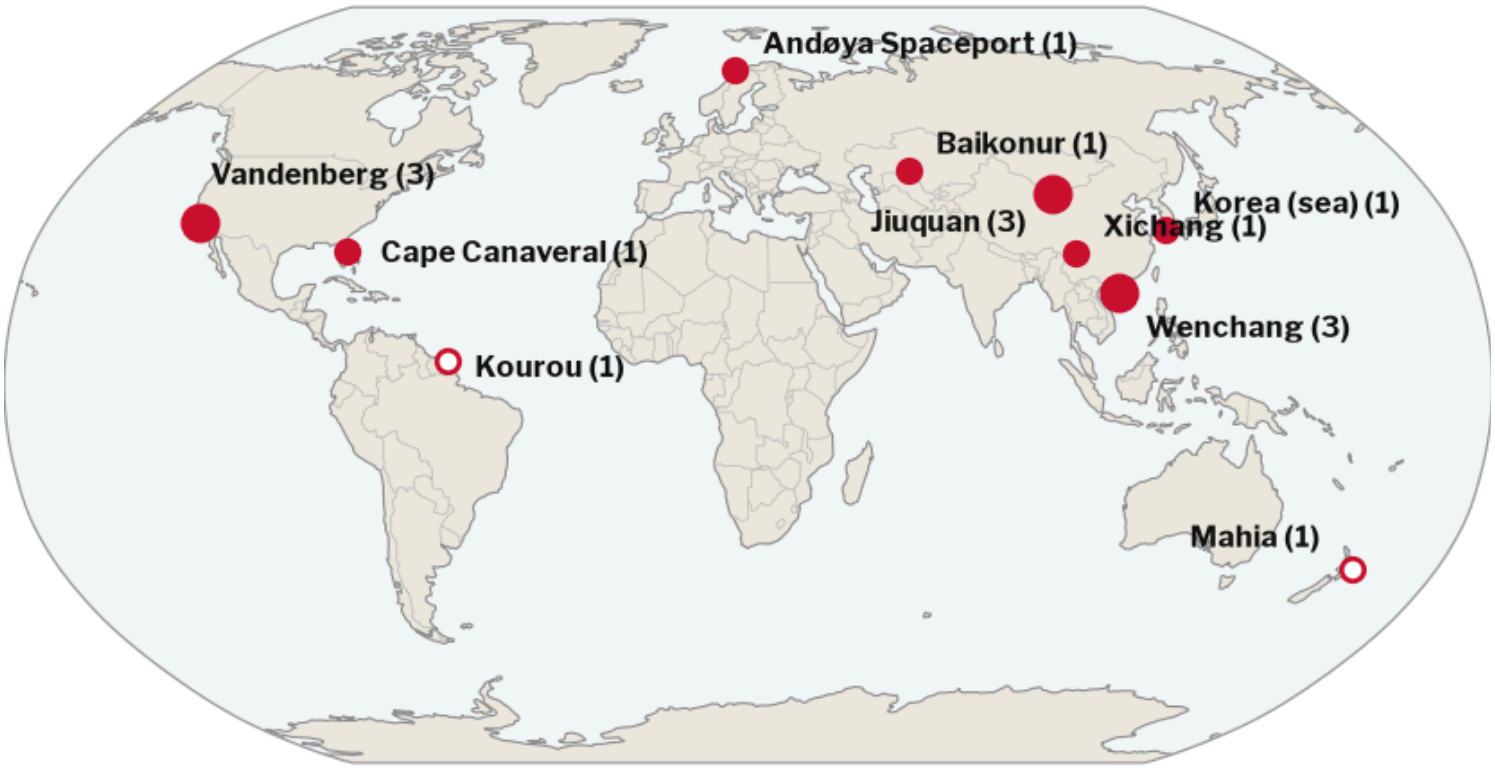
Cygnus NG-25. Sunday, November 29, at 11:00 a.m.; now 10:49 a.m. (Launch Library 2). No reason given.

New to the schedule. USSF-317 on Thursday, November 12; a Long March 2D from Jiuquan on November 11 (Launch Library 2; FAA).

Also listed for November without a specific date in Launch Library 2: Starship Flight 15 from Starbase; Starlink 15-21, NROL-195 and OneWeb 21 on Falcon 9s from Vandenberg; and ISRO's first uncrewed Gaganyaan test flight on a Launch Vehicle Mark 3 and a PSLV-XL with the TDS-01 technology demonstration satellite.

QUICK HIT

Booster for USSF-432. NASASpaceflight lists Falcon 9 first stage B1095 on its eighth flight for Friday's USSF-432, with recovery on Of Course I Still Love You.



Launch sites with orbital launches dated in Launch Library 2 through late November; numbers count launches. Filled: dated launch (Confirmed or Tentative). Hollow: Preliminary. Mahia and Jiuquan also hosted Monday's successful flights.

MAP: THE LAUNCH REPORT · PAD COORDINATES FROM LAUNCH LIBRARY 2 · NATURAL EARTH

III · TRAJECTORIES & HAZARD AREAS

Airspace windows for three Vandenberg Space Force Falcons

The FAA's operations plan lists the times U.S. airspace is set aside; it gives no hazard-area coordinates.

How they fly

Monday's Electron flew an easterly trajectory from Mahia to deliver QPS-SAR-20 to low Earth orbit, and Kinetica 2 left Jiuquan on a southerly path into polar orbit (NASASpaceflight). All three Falcon 9s in Thursday's FAA plan launch from Space Launch Complex 4E at Vandenberg.

Airspace windows

The FAA Air Traffic Control System Command Center's plan for Thursday lists these launch windows, converted from UTC (FAA advisory):

FLIGHT	PRIMARY WINDOW	BACKUP
"R-4" (USSF-432)	**Fri Nov 6, 7:46 p.m.–Sat 12:29 a.m.**	**Sat Nov 7, 7:32 p.m.–Sun 12:15 a.m.**
"R-5" (USSF-627)	Mon Nov 9, 10:03 p.m.–Tue 2:46 a.m.	Tue Nov 10, 9:49 p.m.–Wed 2:32 a.m.
USSF-317	Thu Nov 12, 11:23 p.m.–Fri 4:06 a.m.	Fri Nov 13, 11:09 p.m.–Sat 3:52 a.m.

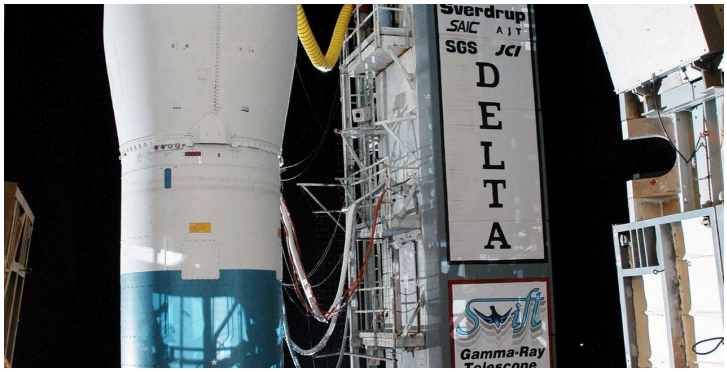
Starlink 15-21, which Monday's plan still carried for Thursday night, is no longer listed (FAA).

IV · REENTRIES & ISS TRAFFIC

Swift comes down over Indonesia; final iROSA spacewalk today

NASA's gamma-ray observatory ended nearly 22 years in orbit Wednesday, and two astronauts were due outside the station this morning.

Neil Gehrels Swift Observatory



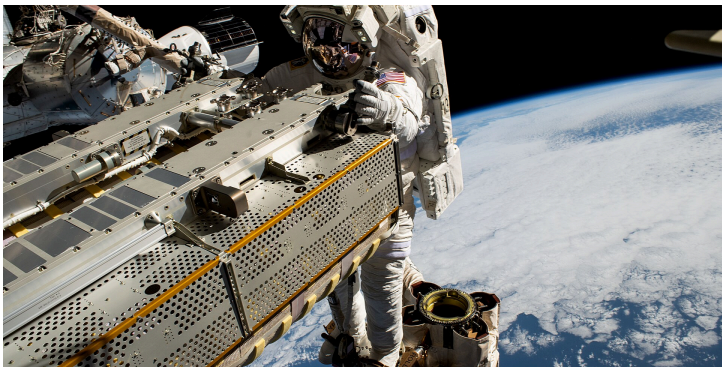
A Boeing Delta II stands ready at Complex 17A, Cape Canaveral, with NASA's Swift spacecraft before its Nov. 20, 2004, launch.

PHOTO: NASA/KSC · PUBLIC DOMAIN

Swift "plummeted back into Earth's atmosphere Wednesday over western Indonesia," Spaceflight Now reported, ending a mission that launched Nov. 20, 2004, on a Delta II. U.S. Space Force tracking, later confirmed by NASA, put reentry at 7:39 a.m. EST Wednesday (12:39 UTC) near 3 degrees north, 99 degrees east, over North Sumatra. NASA said most of the spacecraft was expected to burn up, with a roughly 1-in-2,100 chance that surviving fragments could harm anyone on the ground.

Designed for two years, Swift observed more than 2,000 gamma-ray bursts and more than 1,400 supernovae and contributed to about 7,000 scientific papers. A $30 million Katalyst Space robotic tug intended to raise its orbit lost control after a July 3 launch; NASA called off the capture Aug. 19. Swift made its final observation Oct. 20 (Spaceflight Now).

Station traffic: the last roll-out array



NASA astronaut Woody Hoburg works on a Roll-Out Solar Array outside the International Space Station during a 2023 spacewalk.

PHOTO: NASA · PUBLIC DOMAIN

Canadian Space Agency astronaut Joshua Kutryk and NASA astronaut Anil Menon "will begin a spacewalk expected to last about seven hours" at 8:30 a.m. EST Thursday to "install the final roll-out solar array," the space station account posted this morning. Menon called it "installment 7 of 8," noting last week's array produced a greater than 40 percent power increase on its channel (NASA). Wednesday's Expedition 75 work finished spacewalk preparations (NASA).

The next visitors remain Progress MS-36 at 12:03 a.m. November 24 from Baikonur, China's Shenzhou 24 to Tiangong NET late November, and Cygnus NG-25 at 10:49 a.m. November 29 (Launch Library 2).

Confirmed reentries

CelesTrak's satellite catalog lists these objects as decayed since Monday's issue, in addition to Swift. Dates are UTC.

OBJECT (CATALOG NO.)	LAUNCHED	DECAYED	LAST ORBIT, KM
Fengyun 1C debris (30744)	May 10, 1999	Nov 2	266 × 276
NAPA-1 (46320)	Sep 3, 2020	Nov 2	132 × 149
Starlink-3306 (50163)	Dec 18, 2021	Nov 2	144 × 152
HYPSO-1 (51053)	Jan 13, 2022	Nov 2	150 × 170
Starlink-1327 (45555)	Apr 22, 2020	Nov 3	158 × 161
Starlink-1478 (45755)	Jun 13, 2020	Nov 3	140 × 150
Starlink-30239 (57291)	Jul 10, 2023	Nov 3	196 × 212
Swift (NASA; Space Force)	**Nov 20, 2004**	**Nov 4, 7:39 a.m. EST**	**over North Sumatra**
Cosmos 1408 debris (50058)	Sep 16, 1982	Nov 4	150 × 162

Four Starlinks and the Norwegian HYPSO-1 imaging satellite are among the catalog decays; NAPA-1 was a Thai Earth-observation cubesat (CelesTrak). Swift's catalog record had not yet gained a decay date when this issue closed, so the table uses the Space Force and NASA times reported by Spaceflight Now.

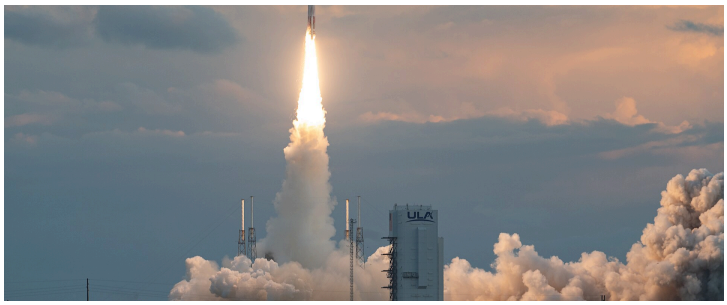
V · INDUSTRY & REGULATION

Vulcan Leo 1 holds Nov. 13; Hubble reboost RFI still open

Rocket Lab's ESCAPADE probes near a Mars burn, ULA's return to flight keeps its date, and Isaacman presses for a controlled Hubble end of life.

Rocket Lab and ESCAPADE

"Get ready: Trans-Mars Injection incoming," Rocket Lab posted Monday about Blue and Gold, the twin Explorer-class probes it built for NASA's ESCAPADE mission with the University of California, Berkeley. "Our two spacecraft … will slingshot Earth and shoot for Mars in four days." A follow-up Tuesday walked through the in-house solar panels, star trackers and reaction wheels as the probes neared the critical burn (Rocket Lab). Separately, Maxwell, a Rocket Lab-built spacecraft for Viasat under NASA's Communications Services Project, returned one of its first on-orbit images and is "healthy and moving through commissioning ahead of schedule" (Rocket Lab).



Vulcan lifts off from Space Launch Complex 41 at Cape Canaveral Space Force Station on its second certification flight on Oct. 4, 2024.

U.S. SPACE FORCE PHOTO BY AIRMAN 1ST CLASS COLLIN WESSON · PUBLIC DOMAIN

United Launch Alliance

Vulcan Leo 1, carrying 40 Amazon Leo satellites on a VC6L with six solid rocket boosters, remains planned for Friday, November 13, "pending range approval" (ULA). Launch Library 2 now carries a 12:54 p.m. EST T-0. The flight is Vulcan's return after nozzle failures on the Northrop Grumman 63XL motors grounded the rocket following the February USSF-87 mission; ULA says the upcoming launch uses an improved nozzle design (Spaceflight Now). A second Vulcan is already stacked for Vulcan Leo 2.

Hubble Request for Information

NASA Administrator Jared Isaacman told NASASpaceflight that an uncontrolled Hubble reentry "would not be the right thing to do," and that any funded visit should reboost the telescope to extend its science return (NASASpaceflight). Hubble is operating on two of six gyroscopes in single-gyro mode, cutting available observation time by about 12 percent; without propulsion it is expected to reenter around 2033–2034. NASA's Request for Information on commercial options for a reboost, servicing or controlled deorbit remains open through the end of November. Isaacman cited the failed Swift rescue as useful experience for any future capture attempt.

QUICK HIT

Part 450 comments closed. The Wednesday deadline for comments on five proposed changes to the commercial launch and reentry licensing rule has passed (docket FAA-2026-9935).

Comment deadlines still open

FCC space licensing. Comments on the Federal Communications Commission's Space Modernization for the 21st Century further notice (SB Docket No. 25-306) are due November 25 and replies December 28 (Federal Register).

Starbase. Scoping comments on up to 456 Starship launches a year at Starbase are due November 30, with meetings November 17 to 19 (Federal Register).

VI · BY THE NUMBERS

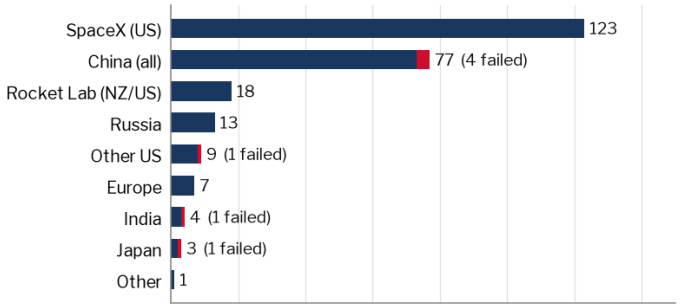
255 orbital launch attempts so far in 2026

Monday's Electron and Kinetica 2 lifted the year-to-date count two above the GCAT October 31 total.

255	248	123	77	18
orbital launch attempts worldwide through Nov 3	reached orbit	by SpaceX, all successful	by China, four failures	by Rocket Lab, all successful

The count

Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) records 253 orbital launch attempts through October 31. Adding Monday's Electron and Kinetica 2 from Launch Library 2 brings the running total to 255 attempts and 248 successes. SpaceX remains at 123, all successful, followed by China with 77, Rocket Lab with 18 and Russia with 13; other U.S. providers have flown nine, Europe seven, India four and Japan three. GCAT counts Japan's October 19 launch of the Martian Moons eXploration probe as a deep-space launch, outside these totals.



Orbital launch attempts in 2026 by provider or country. Navy: success. Red: failure or partial failure. "Other" is South Korea.

CHART: THE LAUNCH REPORT · GCAT THROUGH OCT 31 PLUS MONDAY'S LAUNCHES FROM LAUNCH LIBRARY 2

The failures

According to GCAT, seven launches have failed this year: India's PSLV on January 12; China's Long March 3B on January 16, Ceres-2 on January 17, Tianlong-3 on April 3 and Long March 7A on August 10; Japan's Kairos on March 5; and Blue Origin's New Glenn on April 19, recorded as a partial failure. Galactic Energy's Ceres-2 is listed to fly again Friday (Launch Library 2).

The week ahead

Seven orbital launches have specific dates between Thursday and November 10 (Launch Library 2; FAA): a Long March 2C today, Ceres-2 and USSF-432 on Friday, a Long March 12 on Sunday, USSF-627 on Monday and South Korea's offshore Mir demonstration early Tuesday, with a Long March 2D on Wednesday.

BY THE NUMBERS

99	Electron flights after Monday (Rocket Lab)
18	Spacesail satellites on Kinetica 2 (Launch Library 2)
255	orbital launch attempts in 2026 through Nov. 3 (GCAT; Launch Library 2)
22	years Swift spent in orbit (Spaceflight Now)

Takeaways

1. Electron's 99th launch and Kinetica 2's Spacesail cluster both succeeded Monday.
2. 2026 stands at 255 orbital attempts and 248 successes.
3. USSF-432 slipped to Friday at 7:46 p.m.; Starlink 15-21 is off the near-term FAA plan.
4. Swift reentered Wednesday at 7:39 a.m. EST over North Sumatra.
5. Kutryk and Menon were due to install the station's final roll-out solar array Thursday morning.

KEY DATES AHEAD (EST)

Thu Nov 5	Space station spacewalk, 8:30 a.m.; Long March 2C, 11:00 a.m.
Fri Nov 6	Ceres-2, 6:35 p.m.; USSF-432, 7:46 p.m.
Sun Nov 8	Long March 12, 3:25 p.m.
Mon Nov 9	USSF-627, 10:03 p.m.
Fri Nov 13	Vulcan Leo 1, 12:54 p.m.
Wed Nov 25	FCC space-licensing comments due
Sun Nov 30	Starbase PEIS scoping comments due

SOURCES

Schedules: Launch Library 2 (The Space Devs), Spaceflight Now, NASASpaceflight, FAA operations plan advisory. Agencies: NASA, FAA, FCC, Federal Register, U.S. Space Force. Tracking: CelesTrak, GCAT (Jonathan McDowell). X posts: @RocketLab, @SpaceflightNow, @NASASpaceflight, @Space_Station, @ulalaunch. Photos: NASA and U.S. Space Force (public domain). Maps: Natural Earth (public domain).

The Launch Report is compiled from official agency releases, regulatory filings, orbital tracking data, public posts on X and published reporting. Quotes are reproduced verbatim unless marked as translated or paraphrased. Launch times change often; check with the launch provider before making plans.